

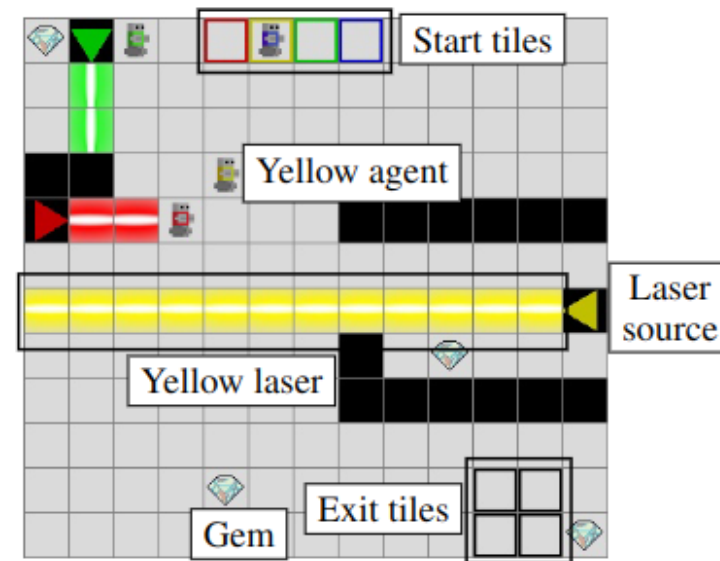


Procedural Generation of Solvable Cooperative Levels for the Laser Learning Environment

Hugo Charels — Master thesis defense

Supervisors: Tom Lenaerts · Yannick Molinghen

ULB — Academic year 2025-2026 · 18/06/2026



Context & motivation

Laser Learning Environment (LLE)

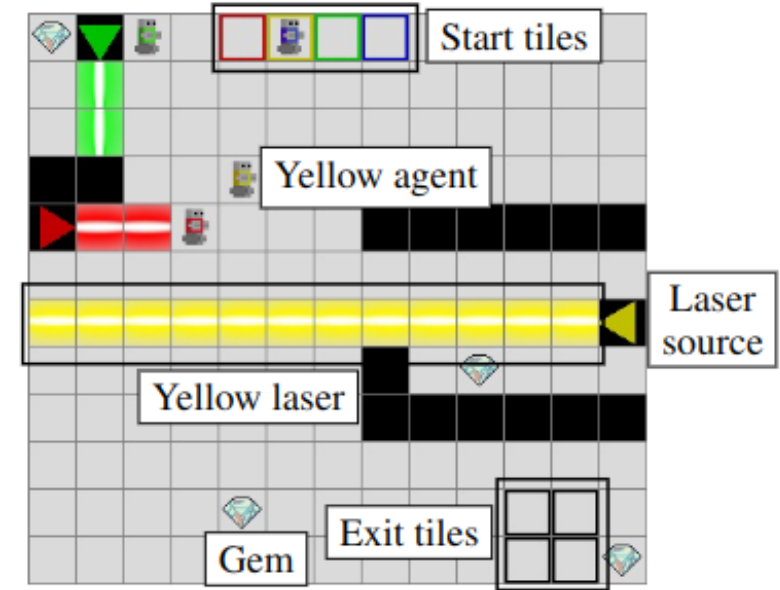
- 2D grid-based fully cooperative multi-agent puzzle
- Each agent must reach its exit tile **simultaneously** with all others
- Laser beams are passable only by an agent of the **matching color**
- Reward signal is sparse: zero credit for intermediate coordination steps

The LLE cooperative mechanic

- A beam blocks **other** colours; each agent is **immune** to its own.
- Block your own beam → free a teammate's path (**locally unrewarded**).

Objective

Generate LLE levels **solvable** and that **require cooperation**.



LLE Level 6 – the hard target



Part 1 – Verification (RQ1–2)

How can we formally verify that a level is solvable and that solving it genuinely requires cooperation?

In this part we present our SAT-based verification tool:

RQ1 – Bounded-horizon solvability

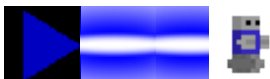
- Formalise the decision problem
- Reduce to SAT: level + horizon $\rightarrow$ Boolean formula
- Verify solvability with a SAT solver

RQ2 – Cooperation detection

- Reuse the SAT encoding, change one rule
- Detect whether cooperation is required
- Classify cooperation structures

Key idea

SAT solver as a verification oracle: given a level and a horizon $T_{\max}$, answer “solvable?” in bounded time.



Bounded-horizon solvability $\rightarrow$ SAT

Decision problem

Given a level L and a horizon $T_{\max}$, does a valid joint trajectory exist?

SAT reduction

Build a Boolean formula $\Phi(L, T_{\max})$. Satisfiable $\Leftrightarrow$ solvable within $T_{\max}$.

$(L, T_{\max}) \rightarrow \boxed{\text{encode } \Phi} \rightarrow \boxed{\text{SAT solver}} \rightarrow \text{SAT / UNSAT} \rightarrow \text{solvable / unsolvable}$

Why SAT? Fast, well-known problem, modern solvers are highly efficient.

Complexity: LLE solvability $\in$ NP and $\leq_p$ SAT. NP-hardness **open**.



Detecting cooperation

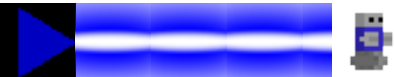
Idea: keep the SAT encoding, change **one** rule: beams no longer block same-colour agents → the cooperation mechanic is **disabled**.

Cooperation criterion

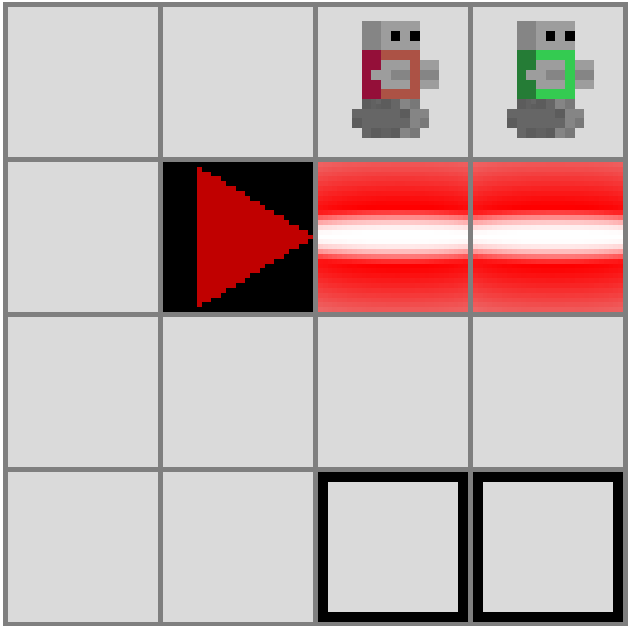
L requires cooperation with $T_{\max} \iff \Phi(L, T_{\max})$ **SAT** and $\Phi_{\text{strict}}(L, T_{\max})$ **UNSAT**.

Three possible outcomes:

Condition	Label	Meaning
Φ UNSAT	UNSOLVABLE	No valid trajectory exists
Φ_{strict} UNSAT	COOPERATIVE	Every solution blocks a beam — cooperation required
Otherwise	NON_COOPERATIVE	A solution exists without the blocking mechanic



Horizon matters



Horizon	Solvable	Coop.
$T_{\max} \leq 2$	no	n/a
$3 \leq T_{\max} \leq 8$	yes	yes
$T_{\max} \geq 9$	yes	no

- **Too small** → agents run out of moves; the level is falsely marked unsolvable
- **Too large** → agents can take longer detours that bypass cooperation requirements

Trade-off

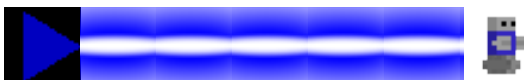
Larger $T_{\max}$ → more actions (more solutions), but bigger formula (longer solve duration). Pick the **smallest** $T_{\max}$ that still admits a solution.



Cooperation profiles

From one SAT model $\rightarrow$ a **dependency graph** (edges: helper $\rightarrow$ beneficiary)

classify by priority: fully coupled $\succ$ mutual $\succ$ distributed $\succ$ chain $\succ$ asymmetric.

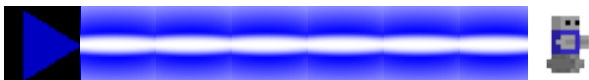
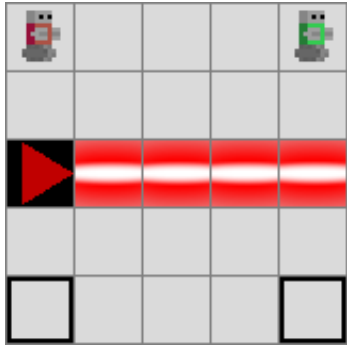


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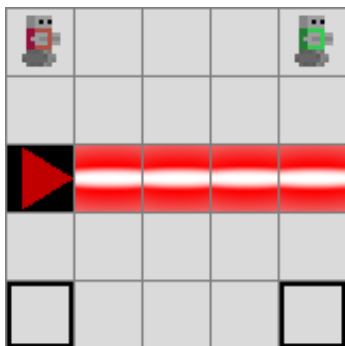


Cooperation profiles

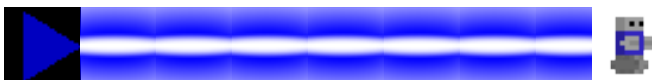
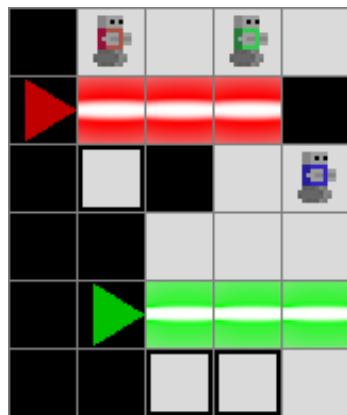
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chain

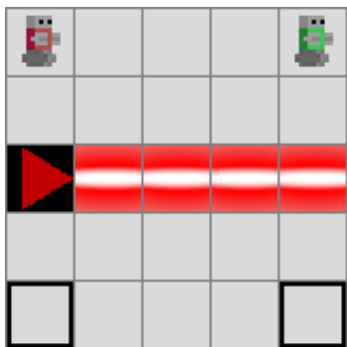


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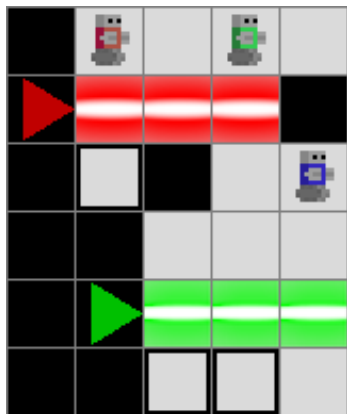
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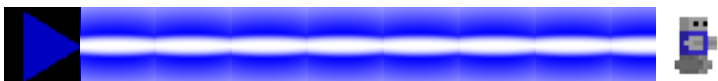
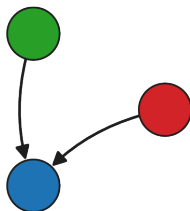
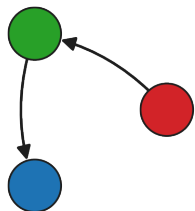
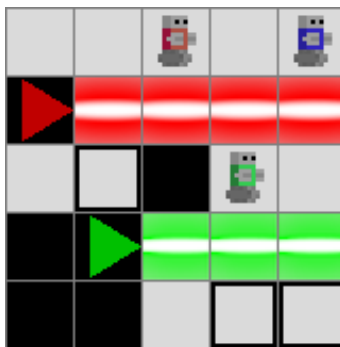
asymmetric



chain



distributed

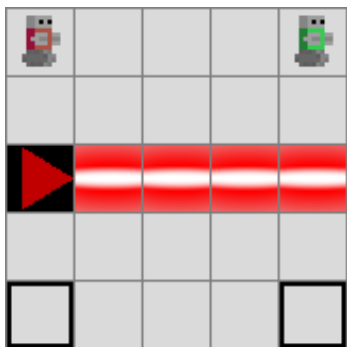


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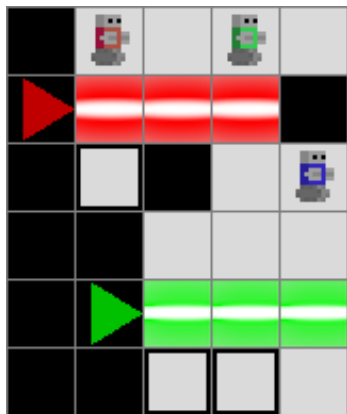
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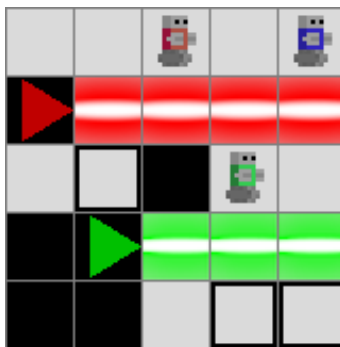
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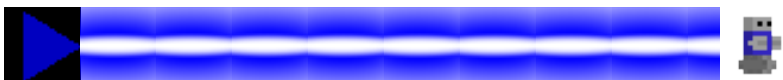
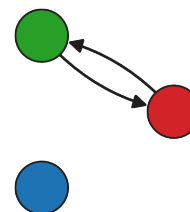
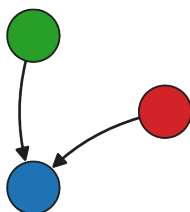
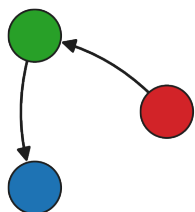
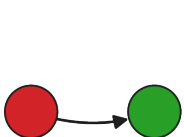
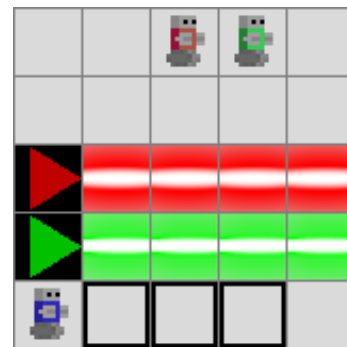
chain



distributed



mutual

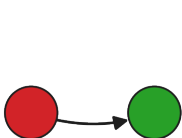
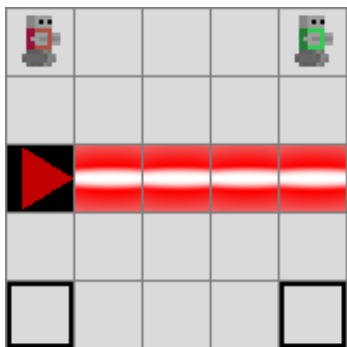


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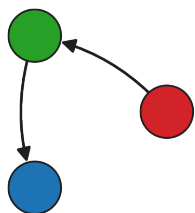
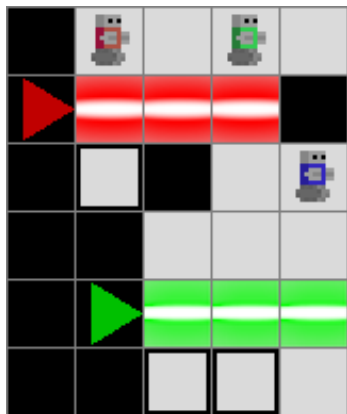
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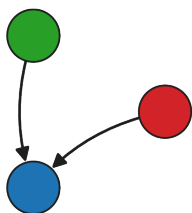
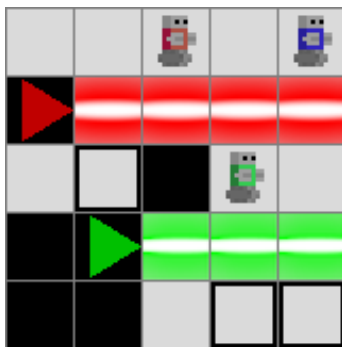
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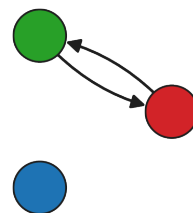
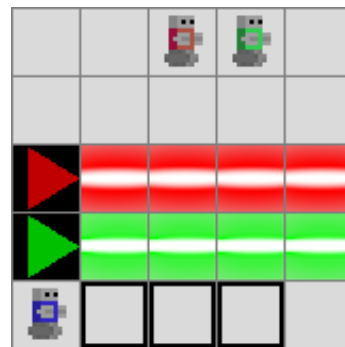
chain



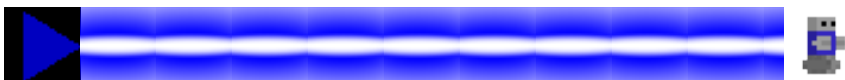
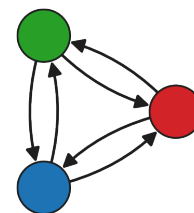
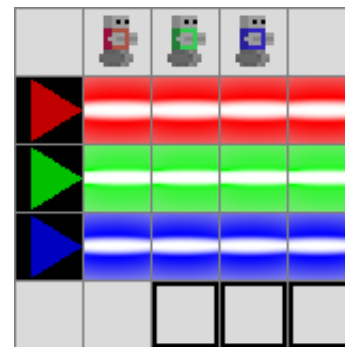
distributed



mutual



fully coupled



Part 2 — Construction (RQ3–4)

How can we embed formal verification inside a procedural generator so that every accepted level is certified, and can we control the cooperation structure by targeting specific profiles?

In this part we present our level generators:

RQ3 — Generator efficiency

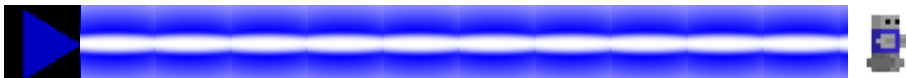
- Random placement with geometric constraints
- Constructive lane-based approach
- Level-6-style clustered design
- SAT oracle filters solvable / cooperative levels

RQ4 — Output diversity

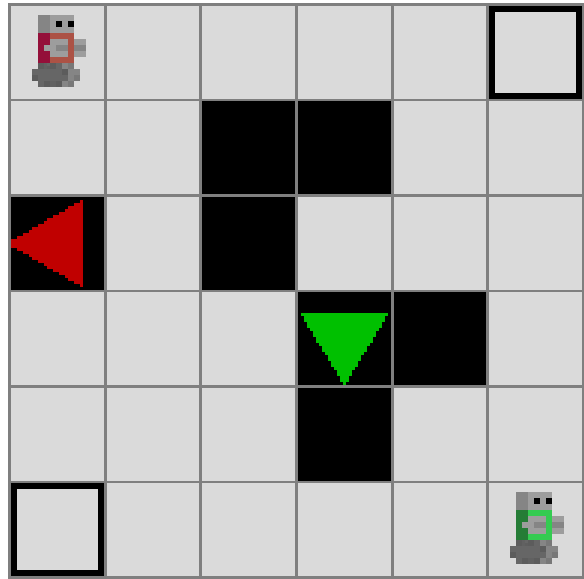
- Five-profile taxonomy of cooperation structures
- Generators can steer towards specific profiles
- Constructive → mutual by design

Key idea

Place-and-check with a SAT oracle: every accepted level is **provably** solvable.

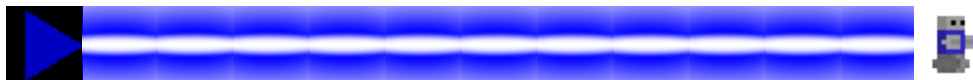


Random Generator

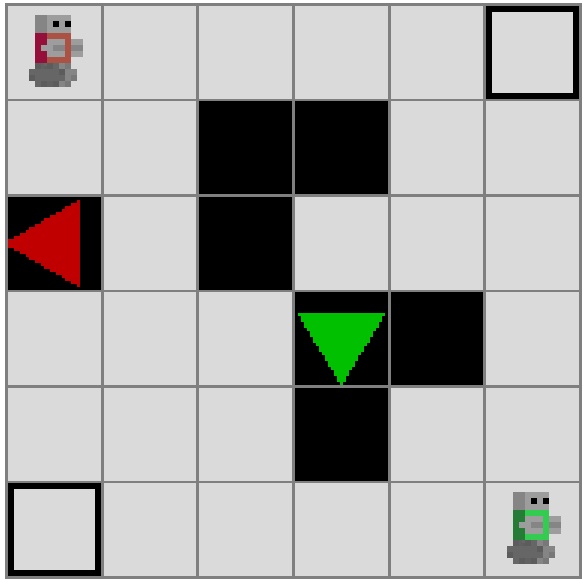


Pure random

Lasers point off-grid, a wall block a laser source

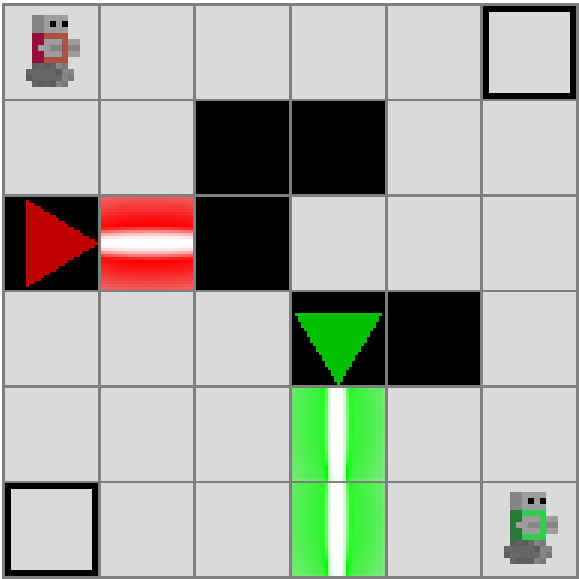


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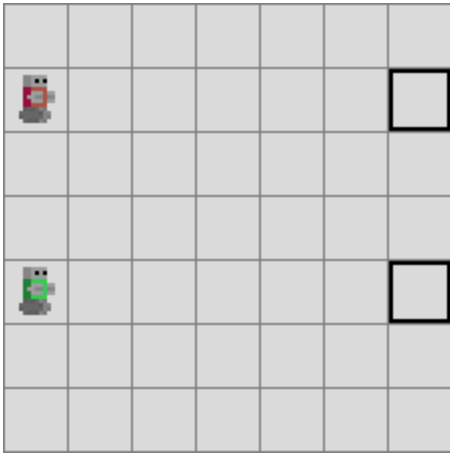
Constrained random

Filters out laser sources that behave like walls.



Constructive Generator

Oriented **lane-based** design. Each agent gets a reserved lane, cooperation arises from crossing lasers. Near-zero rejection rate.

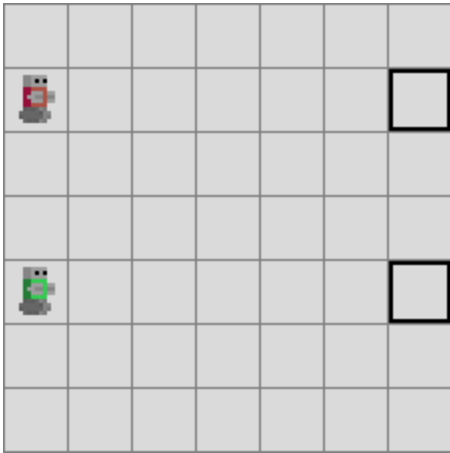


1. Reserve lanes, place agents + exits

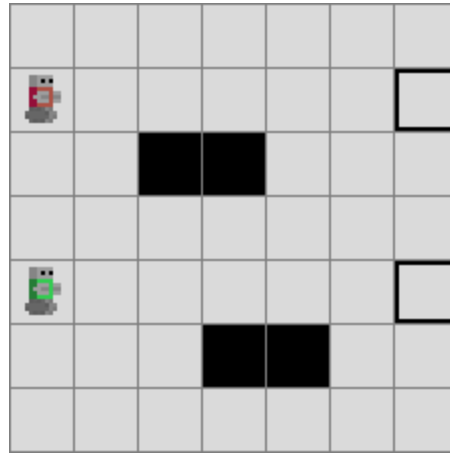


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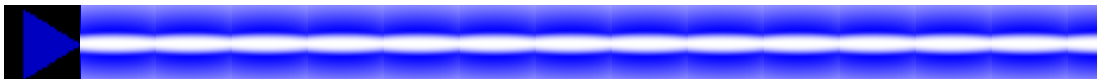
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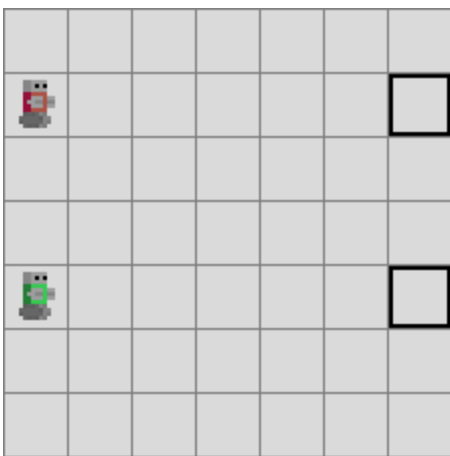


2. Place walls in free cells

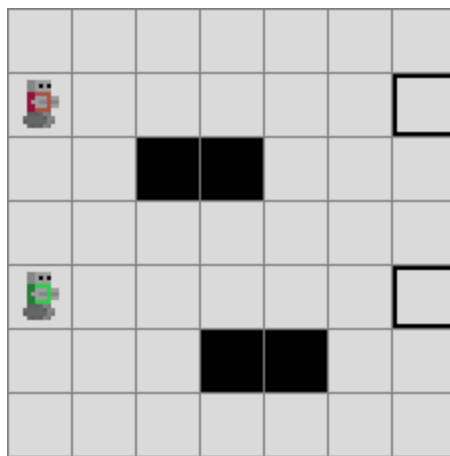


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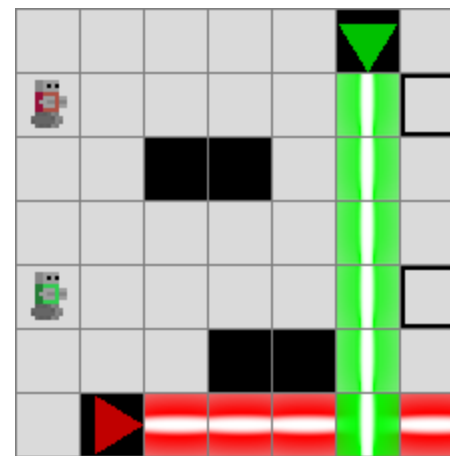
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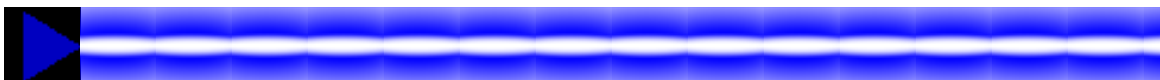
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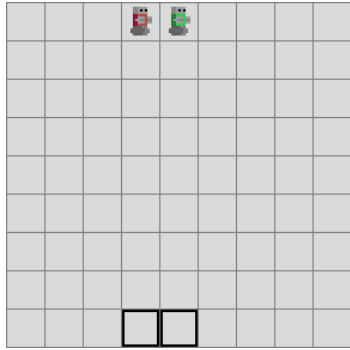


3. Add lasers crossing lanes

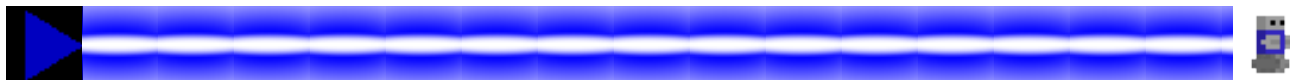


Level-6-Style Generator

Clustered starts + exits on opposite sides. Shared laser corridor = mutual by design. SAT verifies.

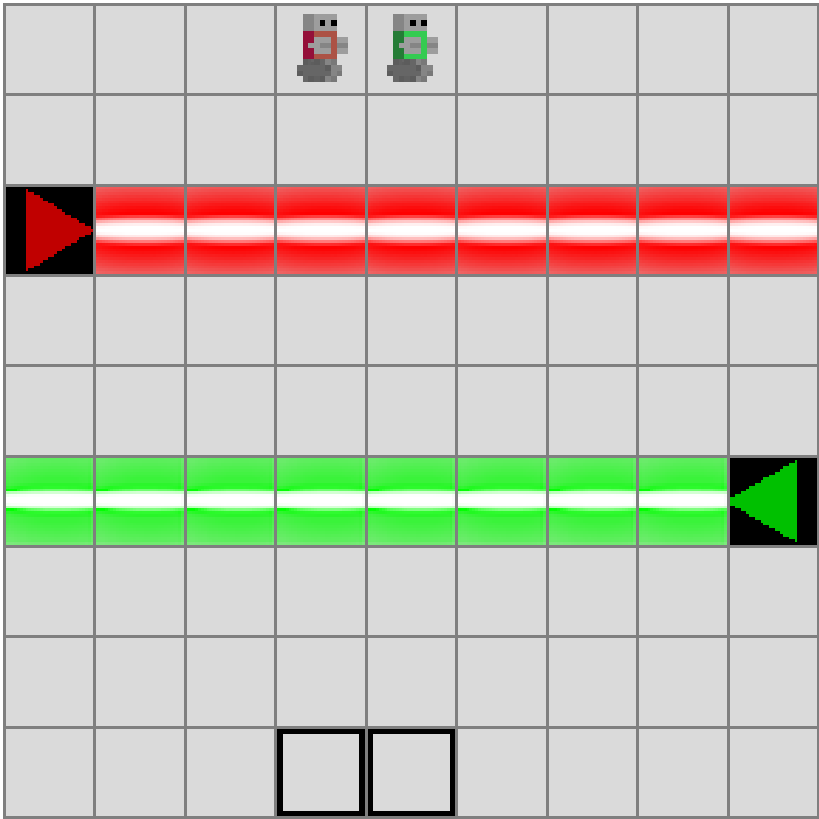
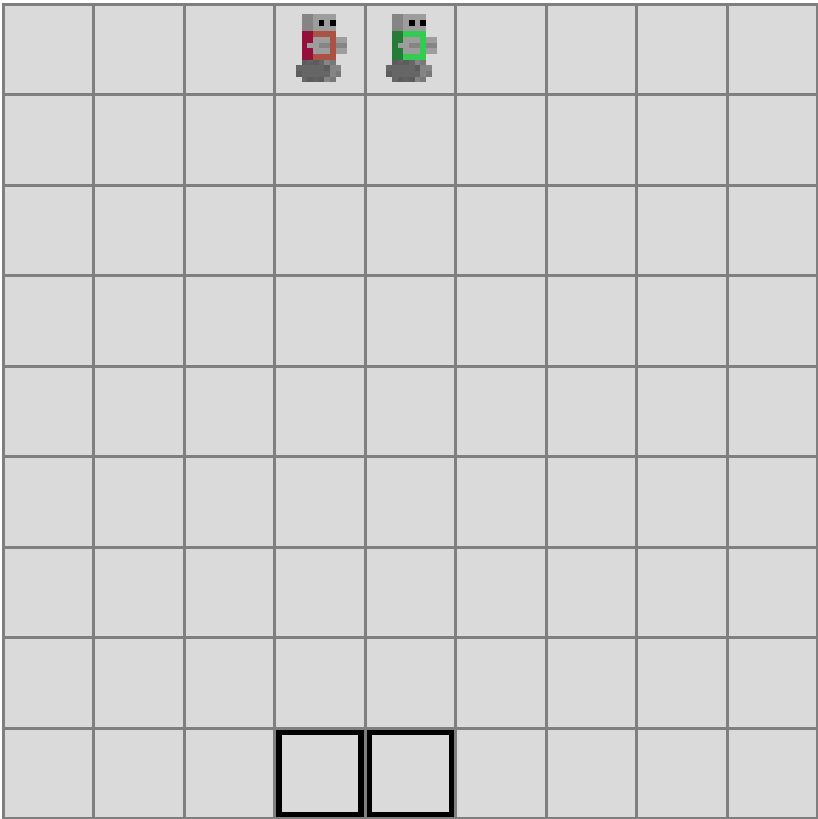


1. Place clusters on opposite sides



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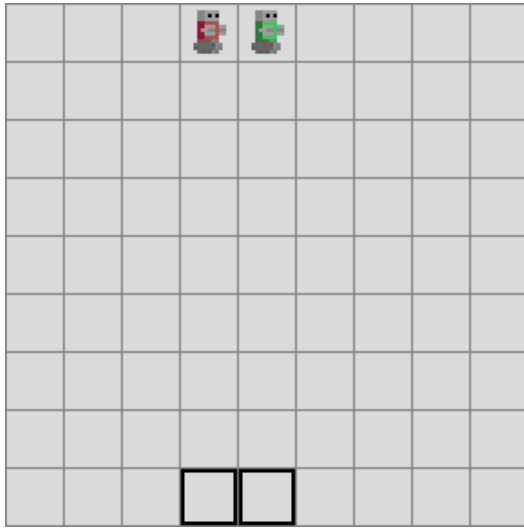
1. Place clusters on opposite sides

2. Fill the central corridor with lasers

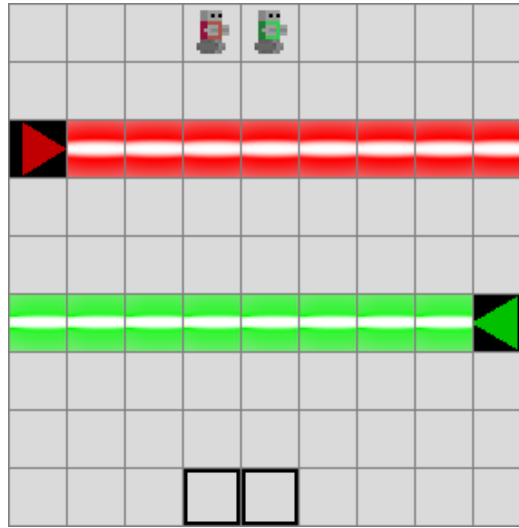


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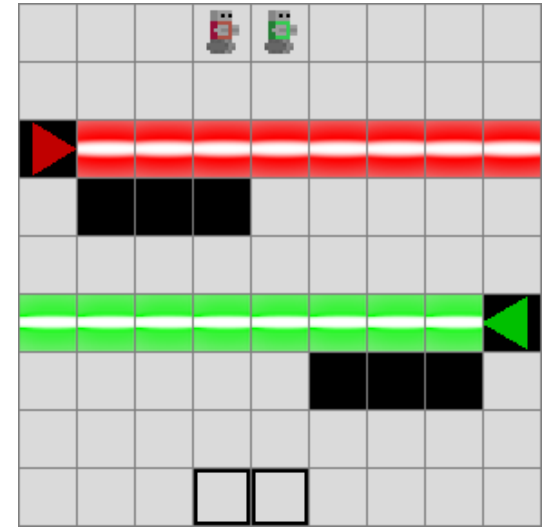
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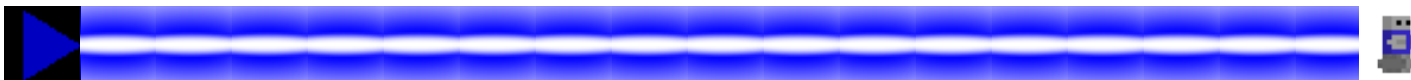
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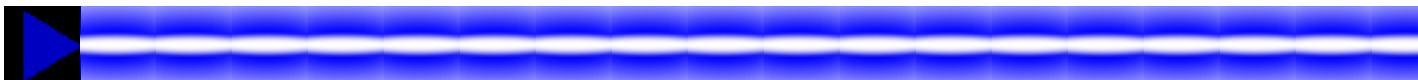
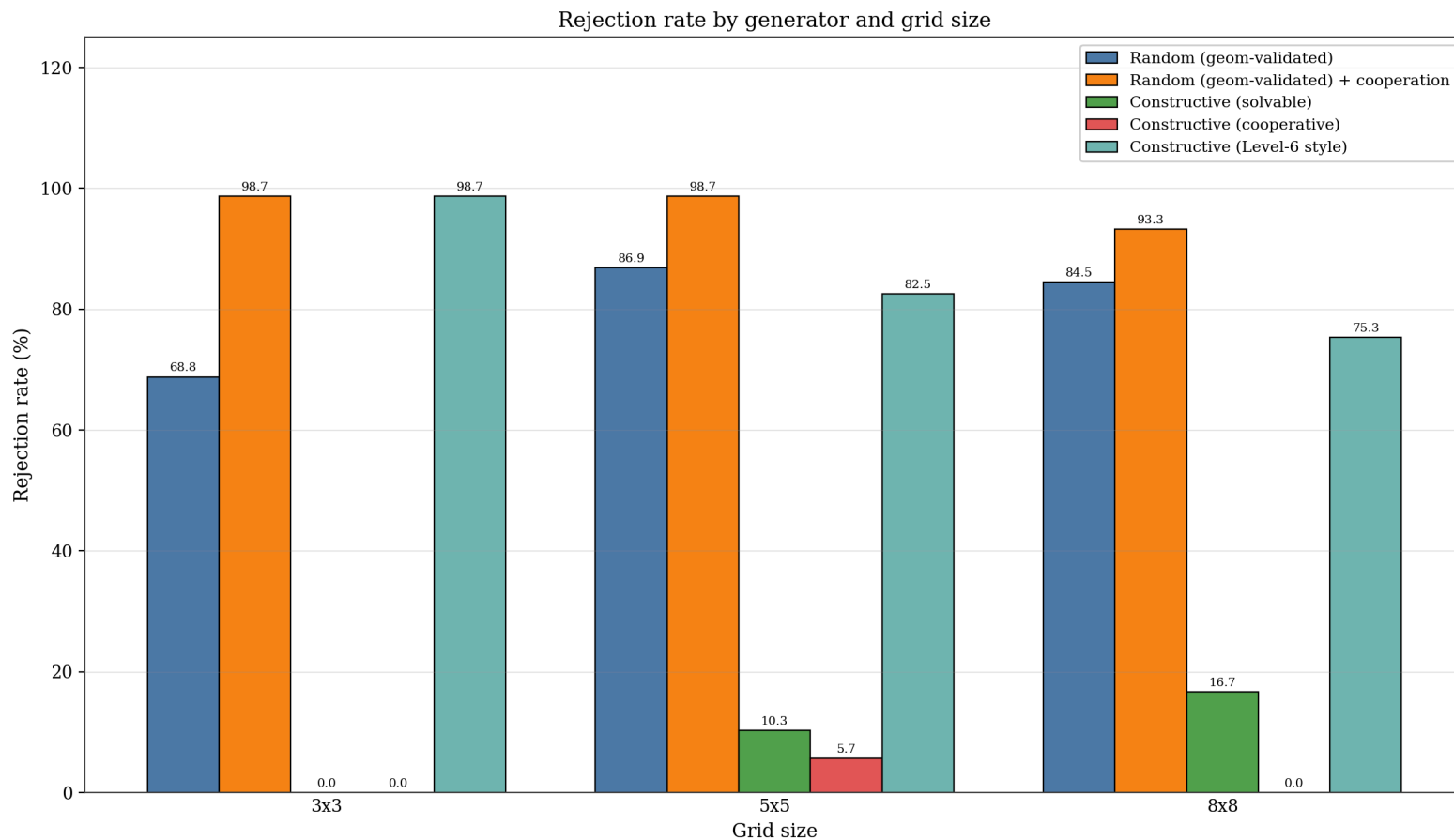
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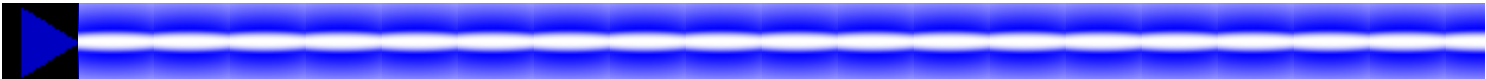
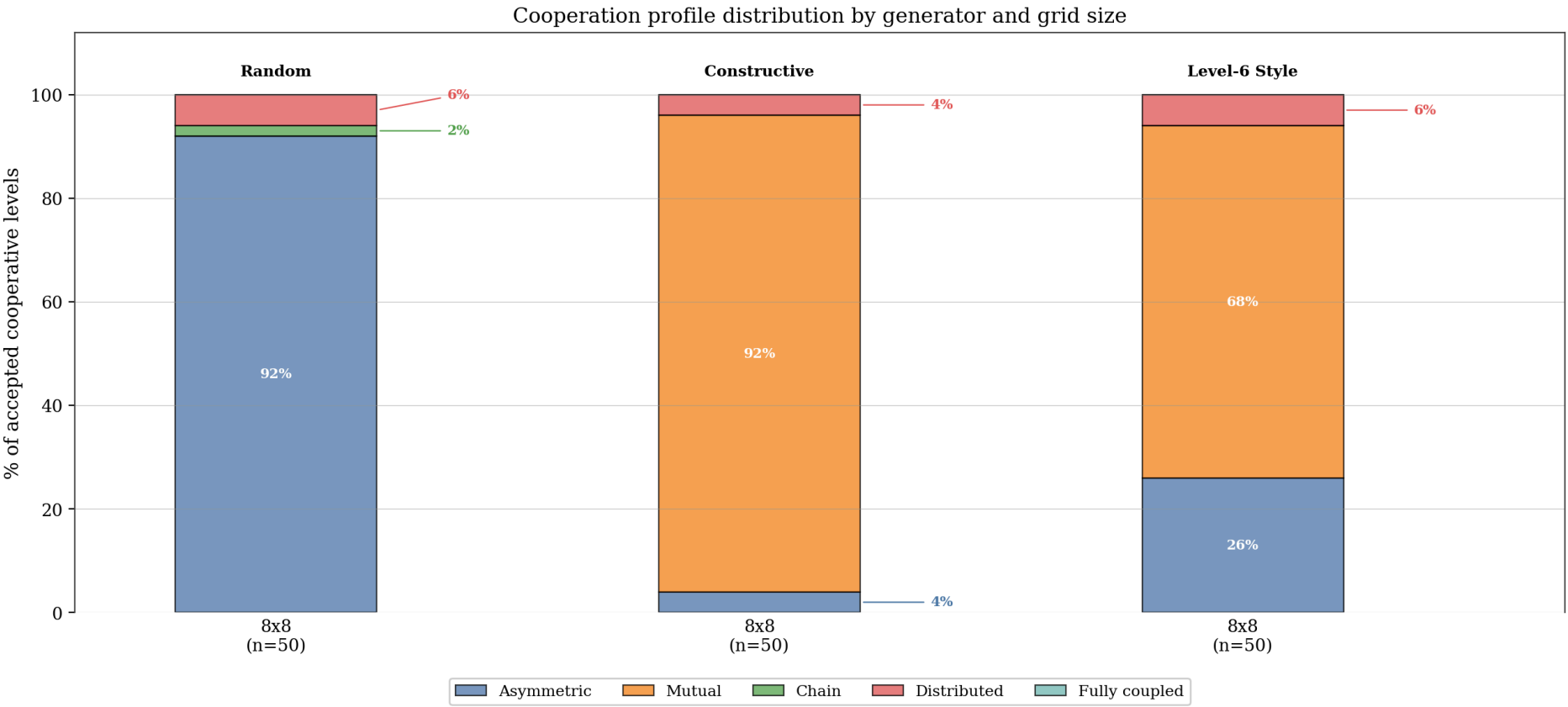
3. Add walls to shape the solution space



Generator efficiency (RQ3)



Output diversity (RQ4)



Part 3 – Experiment (RQ5–6)

Can MARL agents trained on certified levels generalise to held-out ones, and does a staged curriculum outperform direct training on a hard target?

In this part we investigate whether generated levels are learnable:

RQ5 – Learnability

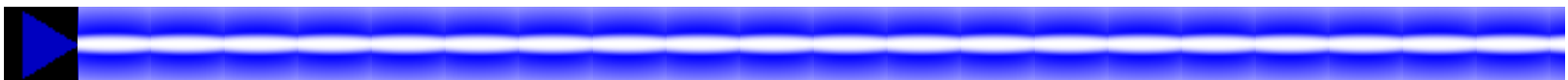
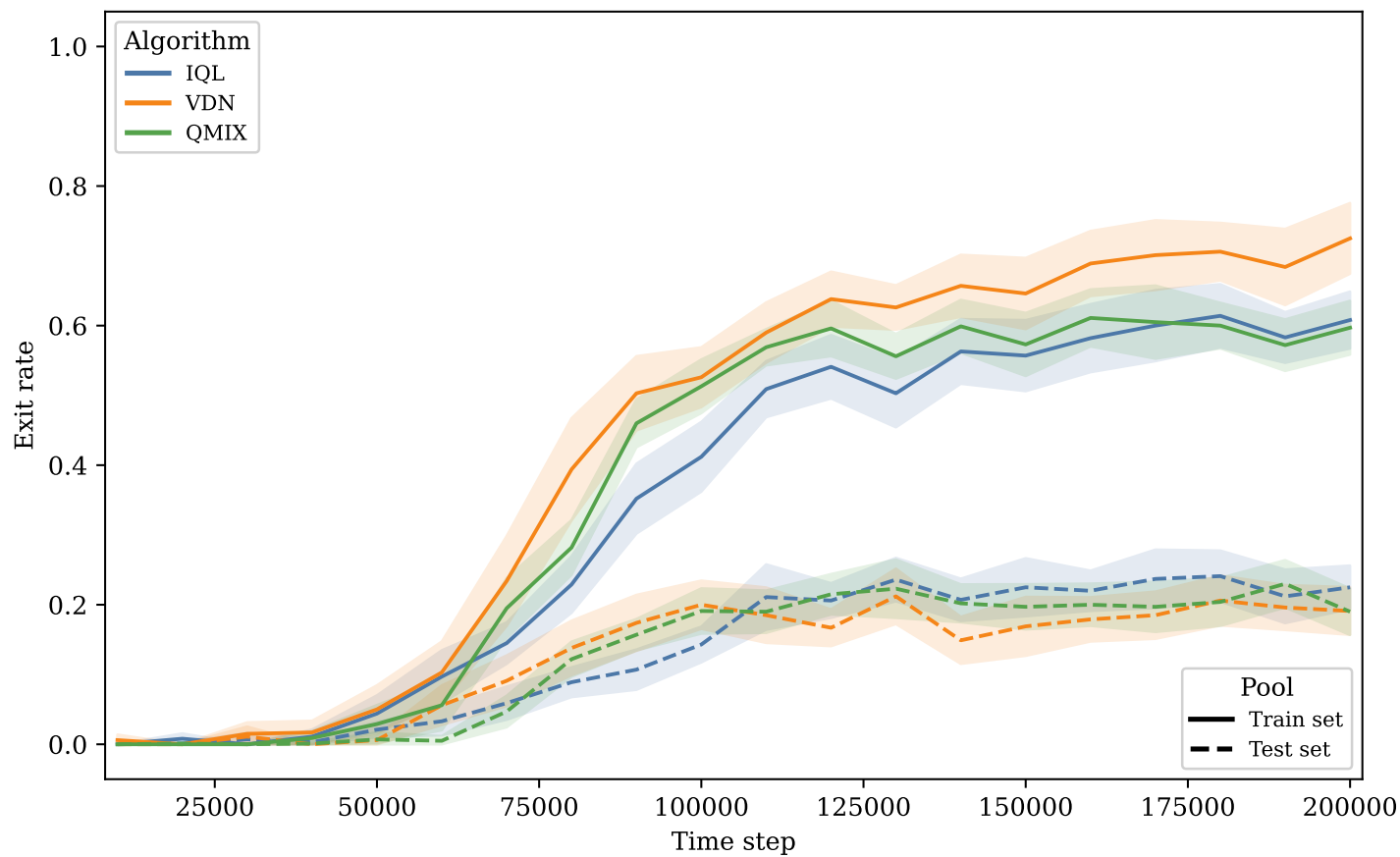
- Train IQL / VDN / QMIX on certified levels
- Measure train vs. test success
- Scale training data to close the generalisation gap

RQ6 – Curriculum learning

- Compare curriculum orderings (direct, forward, mixed, reverse)
- Test harder cooperation targets (mutual, Level-6)



Learnability (RQ5)

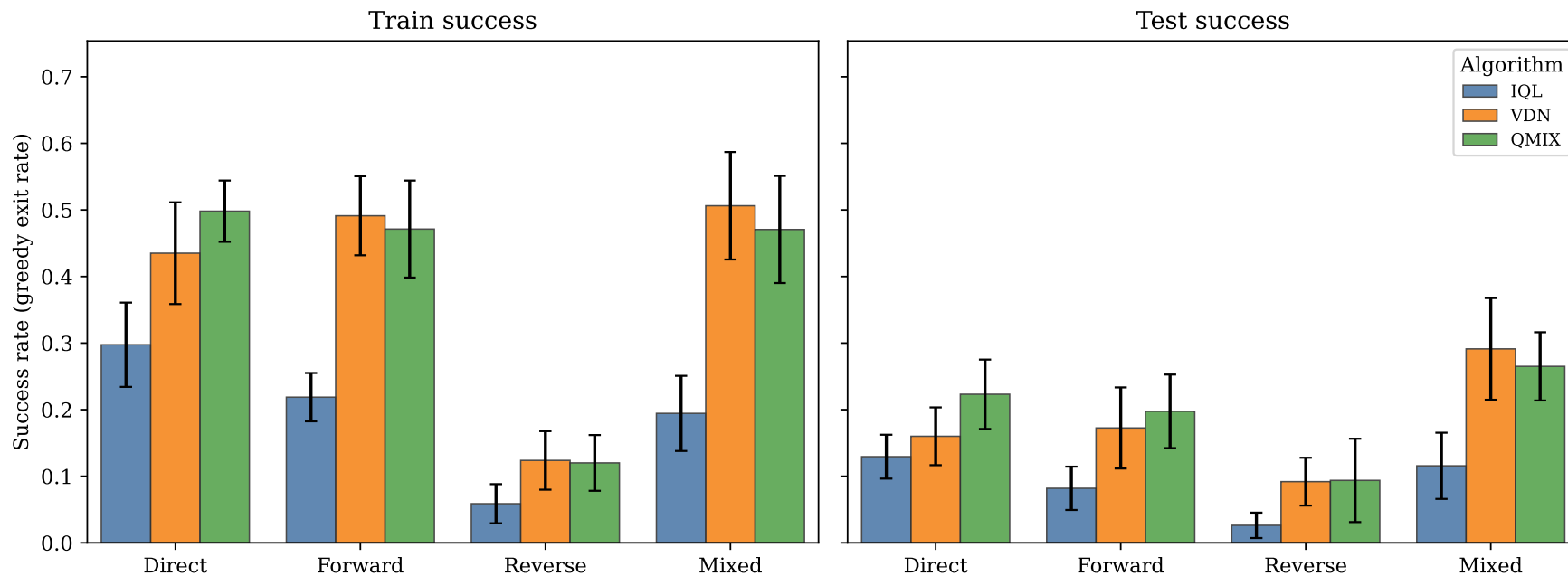


Scaling data (RQ5)

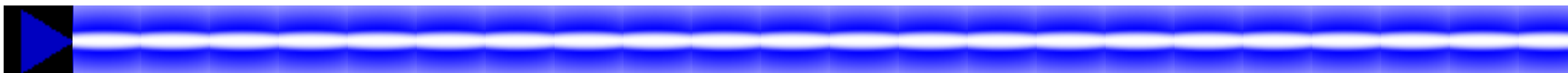


Curriculum ordering (RQ6)

Curriculum strategy on the 6x6/1L target (n=8 seeds, 95% CI)



Four budget-matched schedules (400k steps, 6×6 / 2-agents / 1-laser, asymmetric):



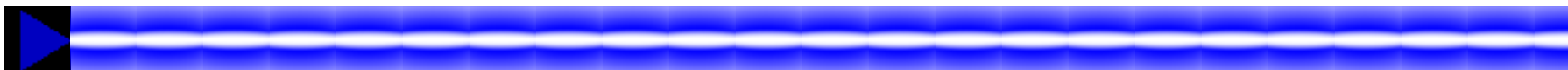
Curriculum targets (RQ6)

Target	Grid	Coop.	Budget	Test
Learnability	5×5	asym	200k	0.20
Curric. order	6×6	asym	400k	0.17
Frontier	6×6	mutual	600k	0.00
Frontier	5×5	mutual	200k	0.00
Level-6	12×13	mutual	2M	0.00

- Reachable target: ordering doesn't beat **direct**; only data **diversity** helps.
- A **cliff** at **mutual** cooperation: success → **zero**, even at 10× budget.

Takeaway

Scoped **negative** result: the base **mutual** task is unlearnable by IQL/VDN/QMIX.



Beyond the thesis: upstream

Solver, cooperation detector, and generators are **merged into the official LLE library** (laser-learning-environment[generator], PyPI, v2.9.0+).

Public API

```
lle.solve • lle.is_cooperative • lle.cooperation_level • lle.generate
```

Takeaway

Not just thesis artefacts — **usable tooling** for the LLE users.



Conclusion

- RQ1 ✓ – SAT verification tool for solvability; in NP, $\leq_p$ SAT.
- RQ2 ✓ – strict counterfactual: coop. $\Leftrightarrow$ SAT $\wedge$ strict-UNSAT.
- RQ3 ✓ – SAT-as-oracle generators; every accepted level **certified**.
- RQ4 ✓ – five-profile taxonomy; generators steer toward mutual.
- RQ5 ✓ – certified levels learnable; scaling data closes the gap.
- RQ6 ✗ – value-based MARL can't learn **mutual** cooperation; no curriculum helps.

Takeaway

Central contribution: procedural generation coupled to **explicit certification**.



Future work

Theory & model

- Settle **NP-hardness** of LLE solvability.
- Add **gems** and true **void** tiles.
- Richer (time-indexed / weighted) cooperation graphs.

Learning

- **Make mutual coordination learnable:** dense / intrinsic rewards.
- Beyond value-based: actor-critics (MADDPG, COMA), MAVEN.
- Surface the *fully coupled* profile.

The certification framework is settled; the open question is **how far it extends**.



Thank you

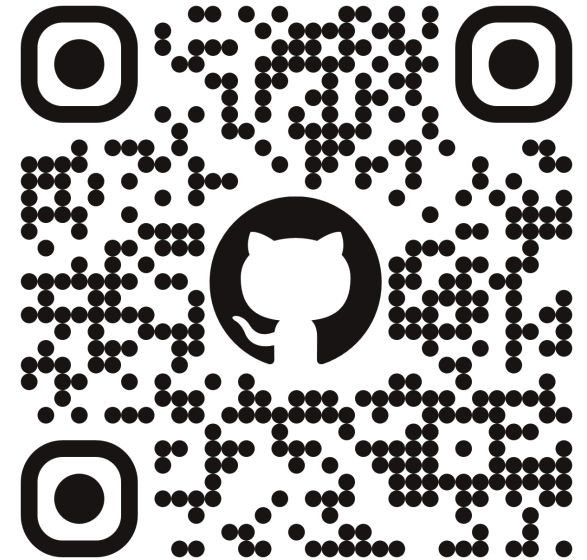
Contributions

1. SAT verification tool for LLE solvability.
2. Cooperation detector + five-profile taxonomy.
3. Solver-in-the-loop certified generators.
4. Merged into the official LLE library.

Key results

1. Local encoding scales; constructive generator $\approx$ free.
2. Scaling certified data closes the generalisation gap.
3. A learnability **cliff** at mutual cooperation.

Questions?



Backup slides

Backup – Full SAT encoding

Variables

- $a_{\{c,x,y,t\}}$: agent c at (x, y) at step t
- $b_{\{c,d,x,y,t\}}$: beam of colour c , direction d , active at (x, y, t)
- $l_{\{c,x,y,t\}}$: laser of colour c active at (x, y, t)

Constraints (4.1–4.13)

Init

- 4.1 agent start tiles
- 4.2 laser-source cells

Movement

- 4.3 forward consistency
- 4.4 global / 4.5 local uniqueness
- 4.6 backward consistency
- 4.7 no overlap / no following
- 4.8 victory • 4.9 stay-on-exit

Lasers

- 4.10 walls block beams
- 4.11 beam propagation
- 4.12 beam $\leftrightarrow$ laser link
- 4.13 no step on foreign laser

$O((n_a + s)p\tau)$ variables; clause count polynomial in n_a, p, s, τ .

Backup – Local vs global uniqueness

Global (A):

$$\bigwedge_c \bigwedge_t \bigwedge_{(x_1,y_1) \neq (x_2,y_2)} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t} \quad O(n_a \tau p^2)$$

Local (B): neighbourhood exclusivity over $\text{next}(x, y)$ + backward consistency

$$O(n_a \tau p)$$

Both admit the **same** satisfying assignments. Local is asymptotically smaller in p but pays a fixed overhead; below the **crossover** grid size the global form is more compact.

Level	3×3	5×5	8×8	Level 6
global	753	8 214	166 736	1 167 640
local	785	6 204	77 516	252 964

Clause counts. Level 6: 7× more clauses than 8×8 yet solves faster — search structure, not size.

Backup – Why the encoding is correct

Decision equivalence

For either movement formulation, $\Phi(L, T_{\max})$ is satisfiable $\Leftrightarrow$ a valid joint trajectory of length $T_{\max}$ exists.

SAT $\Rightarrow$ trajectory. Init fixes one start per agent; forward consistency + uniqueness (global) — or forward + local + backward (local) — force exactly one legal position per agent per step. Movement / collision / laser / exit clauses make the extracted trajectory valid.

Trajectory $\Rightarrow$ SAT. Given a valid trajectory, set a from positions and b, l from the deterministic beam dynamics; every clause family holds by construction.

Consequence. Poly-time, poly-size reduction $\Rightarrow$ bounded-horizon LLE solvability $\in$ NP and $\leq_p$ SAT. NP-hardness open.

Backup – Why the cooperation criterion holds

Cooperation criterion

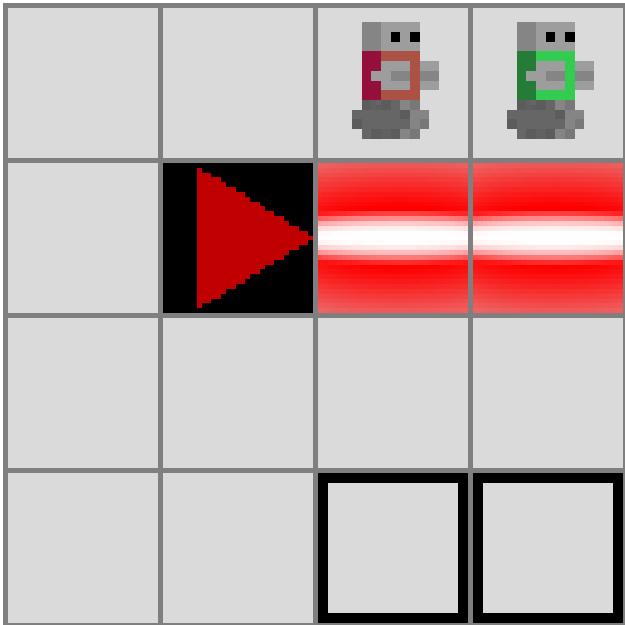
L requires cooperation at $T_{\max} \iff \Phi$ SAT and $\Phi_{\{\text{strict}\}}$ UNSAT.

($\Rightarrow$) Cooperation $\Rightarrow$ solvable, so Φ SAT. If $\Phi_{\{\text{strict}\}}$ were SAT, the strict trajectory would be a valid standard solution **without** blocking – contradiction.

($\Leftarrow$) Φ SAT $\Rightarrow$ solvable. If some standard solution used **no** blocking, both semantics give identical beams along it, so it would satisfy $\Phi_{\{\text{strict}\}}$ – contradicting UNSAT. Hence every solution blocks at least one beam.

Strict encoding changes one clause family: $b_{c,d,x',y',t} \leftrightarrow b_{c,d,x,y,t}$ instead of $b_{c,d,x',y',t} \leftrightarrow (b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t})$.

Backup – Horizon-dependence of cooperation

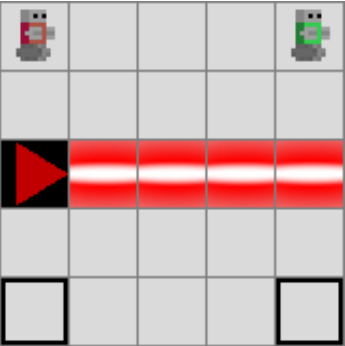


Same level, three labels depending on $T_{\max}$:

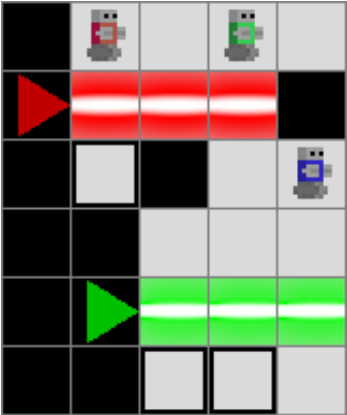
Horizon	Solvable	Coop.
$T_{\max} \leq 2$	no	n/a
$3 \leq T_{\max} \leq 8$	yes	yes
$T_{\max} \geq 9$	yes	no

Too generous $\Rightarrow$ a **detour** fits $\Rightarrow$ under-detected. Too tight $\Rightarrow$ rejected as unsolvable. We pick the smallest horizon fitting a short solution + margin.

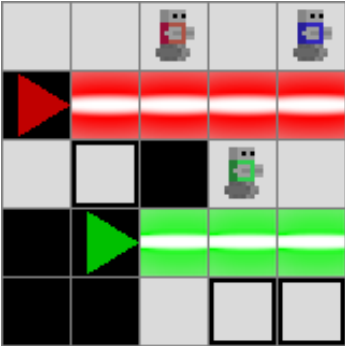
Backup – Cooperation profiles in detail



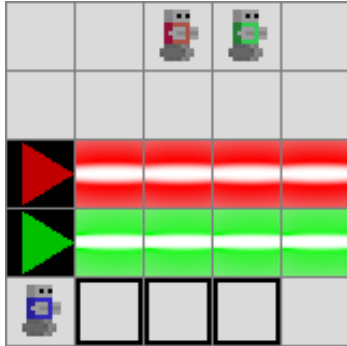
asymmetric



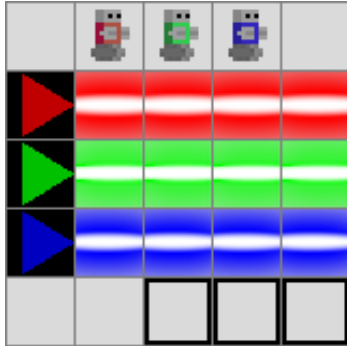
chain



distributed



mutual



fully coupled

